

# Artificial Intelligence: Technical Definition & Framework

Standard Reference & Canonical Taxonomy Specification | REV 4.2

|                |                |                    |                  |
|----------------|----------------|--------------------|------------------|
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## 1. FORMAL DEFINITION

Artificial Intelligence (AI) is defined as the capability of an engineered system to acquire, process, and apply structured knowledge to perform tasks typically requiring cognitive functions. It encompasses algorithms that emulate perception, reasoning, learning, and autonomous decision-making to achieve explicit objectives within defined operational environments.

## 2. ARCHITECTURAL PILLARS

|                                                                                                       |                                                                                                           |
|-------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| <b>1. Perception</b><br>Processing sensor input, computer vision, acoustics, and NLP representations. | <b>2. Reasoning</b><br>Symbolic logic, probabilistic inference, automated planning, and knowledge graphs. |
| <b>3. Learning</b><br>Supervised, unsupervised, and reinforcement parameter optimization.             | <b>4. Autonomy</b><br>Closed-loop control, goal formulation, dynamic pathing, and self-correction.        |

## 3. SUBFIELD TAXONOMY MATRIX

| SUBFIELD               | PRIMARY MECHANISM                   | KEY METRIC           |
|------------------------|-------------------------------------|----------------------|
| Machine Learning (ML)  | Statistical parameter optimization  | Generalization Error |
| Deep Learning (DL)     | Multi-layer representation learning | Feature Abstraction  |
| Natural Language (NLP) | Transformer self-attention models   | Perplexity / BLEU    |
| Computer Vision (CV)   | Convolutional & spatial embeddings  | mAP / IoU            |

## 4. INTELLIGENCE SPECTRUM

- Artificial Narrow Intelligence (ANI): Task-specific optimization operating within defined boundaries.
- Artificial General Intelligence (AGI): Cross-domain cognitive adaptability matching human versatility.
- Artificial Superintelligence (ASI): Theoretical operational capacity exceeding collective human capability.